TRACKING THE RISE OF AI/ML IN OPEN-SOURCE DEVELOPMENT AT BIG-DATA SCALE DATASET: GITHUB ARCHIVE (GH ARCHIVE)

AUTHORS

Arushi Nirmal & Yixiu Liu ·

CS 131 Final Project ·

GITHUB

https://github.com/arushiSH5-cyber/CS-131-FINAL-PROJECT-

INTRODUCTION

AI/ML has reshaped software over the last five years. GitHub — the world's largest open-source platform — records every push, PR, issue, star and fork publicly. By measuring this activity at scale we can see, with real data, how fast the AI/ML ecosystem is actually growing versus software as a whole.

RESEARCH QUESITON

Are AI/ML repositories growing faster than general software repositories from 2021–2026, measured by event volume, event mix (push / PR / issue / star / fork), and new-repo creation?

DATASET

GH Archive — an hourly archive of the public GitHub event timeline. Each hour is one gzip-compressed newline-delimited JSON file (https://data.gharchive.org/YYYY-MM-DD-H.json.gz), one JSON object per event (PushEvent, PullRequestEvent, IssuesEvent, ForkEvent, WatchEvent, …), each carrying type, actor, repo, payload, created_at.

PHASE 1 - PROFILING

EXPLANATION

du / wc — How big? How many rows?

- *time ls *.json.gz | wc -l (0.006 s)*
- *Output: 552 files*
- *time du -ch *.json.gz | tail -1 (0.006 s)*
- *Output: 35 GB compressed*
- *time zcat *.json.gz | wc -c (38m 29.020s)*
- *Output: 273,494,892,306 bytes, or approximately 273.49 GB uncompressed*
- *time zcat *.json.gz | wc -l (38m 37.140s)*
- *Output: 85,339,282 GitHub events*
- *Ten-core parallel byte count (13m 26.302s)*
- *Output: the same 273,494,892,306 bytes, with a 2.9× speedup.*

head / tail / cut / column — Peek at the schema

- *time zcat 2024-01-15-15.json.gz | head -1 | jq . (0.013 s)*
- *Identified the main fields: id, type, actor, repo, payload, public, and created_at.*
- *time zcat 2024-01-15-15.json.gz | tail -1 | jq -c ... (9.290 s)*
- *tail was slower because gzip had to decompress the entire hourly file.*
- *time ... | cut -d',' -f2 | sort | uniq -c | sort -nr (19.560 s)*
- *Extracted and counted event types from one hourly file.*
- *time ... | sort | uniq -c | sort -nr | column -t (17.718 s)*
- *Displayed the event counts in an aligned table.*

sort | uniq -c | sort -nr — Top categories and cardinality

  - *Event-type distribution command (30m 4.864s)PushEvent: 52,244,295, or 61.2%*
  - *CreateEvent: 10,597,330*
  - *PullRequestEvent: 6,005,006*
- *Distinct-actor count command (30m 14.693s)*
- *Output: 8,268,278 distinct actors*
- *Distinct-repository count command (29m 30.237s)*
- *Output: 13,206,596 distinct repositories.*

awk — Numeric aggregate

  - *Average commits-per-push command (29m 30.442s)Total commits: 159,502,273*
  - *Push events: 52,244,295*
  - *Average: 3.053 commits per push.*

grep -c — Filtering counts

- *AI/ML keyword filtering command (17m 43.542s)*
- *Output: 787,912 matching events, approximately 0.92% of all activity.*
- *PullRequestEvent filtering command (12m 29.191s)*
- *Output: 6,005,006 events, exactly matching the independent jq count.*

PHASE 2 — BREAKING

EXPLANATION

8 Hours data is big enough to let Excel and pandas braked

In-memory tools fail BEFORE the analysis even starts — Excel silently throws away 35% of the rows, pandas dies just reading the bytes — because both must hold everything at once.

Streaming (constant memory) and distributed (bounded memory per core) tools finish the identical aggregation in minutes/seconds.

This spreadsheet may look different.

Here's a list of what changed when you opened it in Numbers.

Your data has 1,612,733 rows. Tables in Numbers can't support more than 1,000,000 rows. Some content was not imported.

Sheet: Sheet 1 (Table: excel_rowcap)

```
ishigaki-cs6@spark-833c:~/final_project/2_breaking$ cd ~/final_project
( ulimit -v 8000000    # cap THIS subshell at ~8 GB virtual memory
  .venv/bin/python -c 'import pandas as pd
df = pd.read_json("data/day.json", lines=True)
print(df.shape)' )
Traceback (most recent call last):
  File "<string>", line 2, in <module>
  File "/home/ishigaki-cs6/final_project/.venv/lib/python3.12/site-packages/pandas/io/json/_json.py", line 829, in read_json
    json_reader = JsonReader(
                  ^^^^^^^^^^^
  File "/home/ishigaki-cs6/final_project/.venv/lib/python3.12/site-packages/pandas/io/json/_json.py", line 936, in __init__
    self.data = data.read()
                ^^^^^^^^^^^
MemoryError
```

ONE DAY – 2024-01-15, 5,528,301 events, 18.9 GB raw JSON:

Method	Tool	Wall-clock	Peak memory	Completed?
Load-into-memory	pandas (8 GB cap)	(dies)	8 GB cap hit	NO – MemoryError
Load-into-memory	pandas (uncapped)	–	~270 GB needed	not attempted – extrapolated (14.3x)
Streaming	jq | sort | uniq -c	5m34.8s	~MB, constant	yes
Distributed	PySpark local[*] (20c)	26.0s	bounded/core	yes

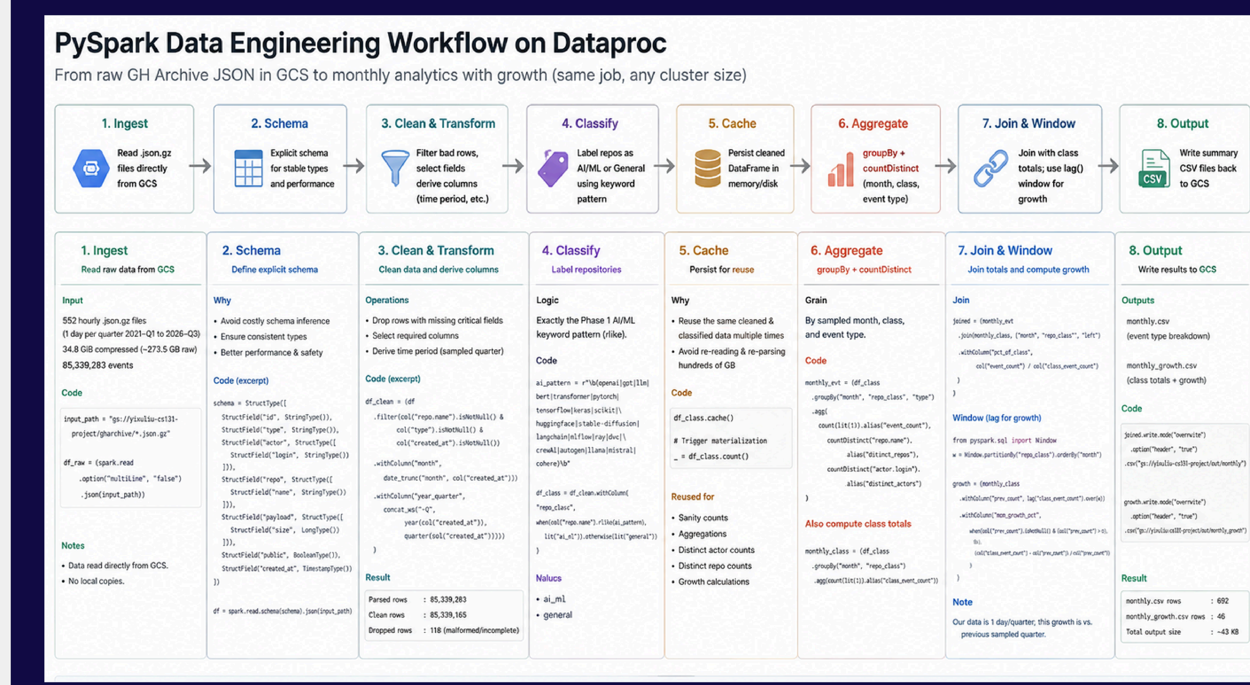
FULL DATASET – 23 days, 85,339,283 events, 37.3 GB gz (~273.5 GB raw):

Method	Tool	Wall-clock	Completed?
Load-into-memory	pandas	–	NO – would need ~3.9 TB RAM
Streaming	zcat | jq | uniq -c	see 1_profile/profiling.txt (same command over all 552 files)	
Distributed	PySpark local[*] (20c)	200.4s	yes

PHASE 3— SCALING

Experiment setup

- *Processed 85.3 million GitHub events from 552 gzip JSON files*
- *Raw data remained in Google Cloud Storage and was read directly through gs://*
- *Used the same PySpark job for every experiment*
- *Tested three compute configurations:*
  - *Single-node: 4 shared vCPUs*
  - *2 workers: 8 worker vCPUs*
  - *4 workers: 16 worker vCPUs*
- *Job included schema definition, filtering, classification, groupBy, joins, window functions, and caching*



Cluster	Worker vCPUs	Job RUNNING->DONE	Speedup	Efficiency	End-to-end `time`
single-node	4 (shared)	2617 s (43m37s)	1.00x	–	43m49.6s
2 workers	8	1048 s (17m28s)	2.50x	1.25	17m37.2s
4 workers	16	621 s (10m21s)	4.21x	1.05	10m30.5s

PHASE 4— ANALYSIS

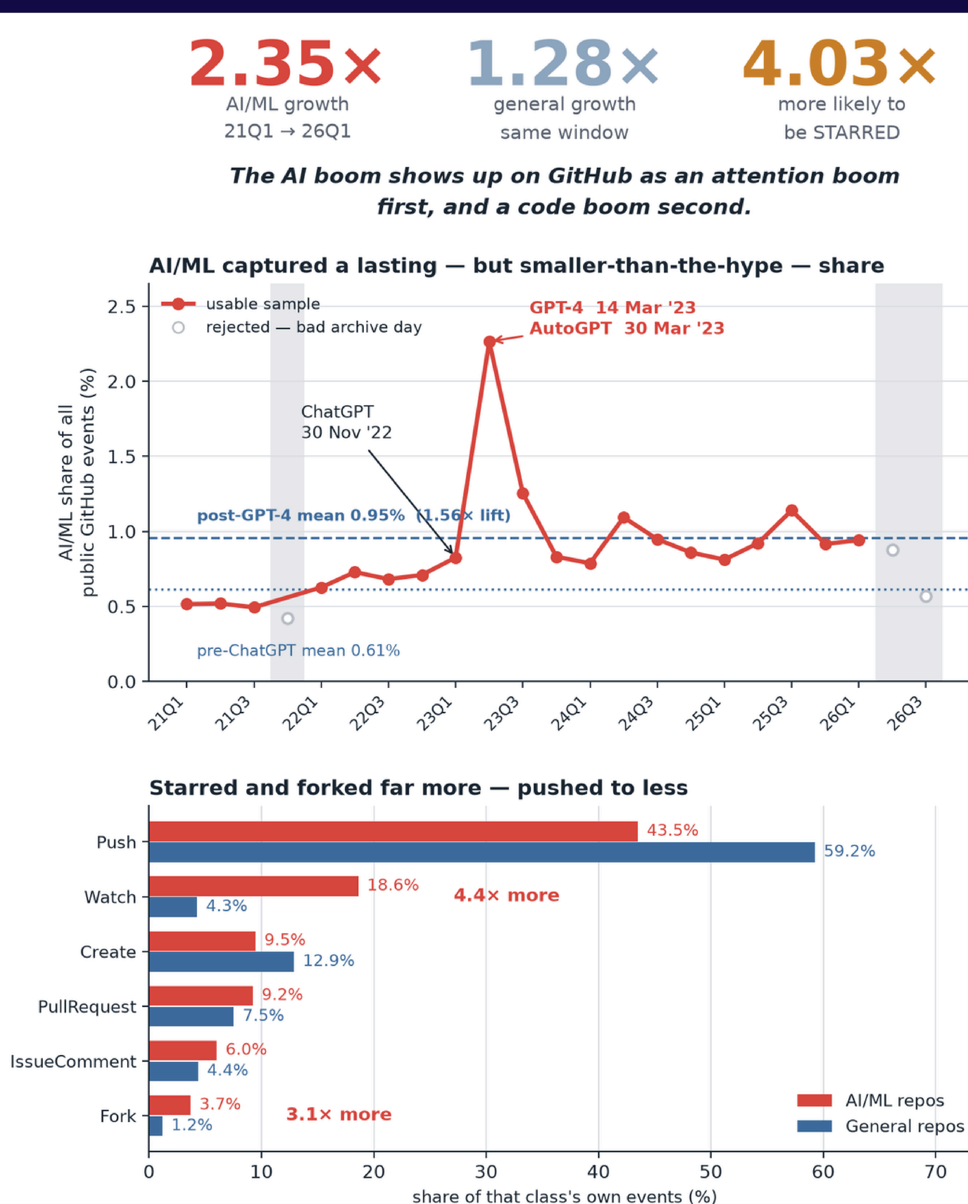
EXPLANATION

Loaded Phase 3's aggregated CSVs (43 KB from 273.5 GB) into pandas and rejected 3 of 23 sampled days first — 21Q4 had only 32% of a median day's events, 26Q2/Q3 lost stars and PRs upstream — so the −68.65% in the growth CSV is an archive artifact, not a real collapse.

AI/ML activity grew 2.35× from 2021 to 2026 vs 1.28× for general software (18.7%/yr vs 5.1%/yr) — 3.7× faster, but still under 1% of all public GitHub events. Share peaked at 2.26% in 2023-Q2, the GPT-4 + AutoGPT quarter, when distinct AI-repo actors jumped 3,734 → 19,977 in three months. Most left; a 1.56× permanent lift remains.

The durable difference isn't volume — it's attention. AI/ML repos are starred 4.4× and forked 3.1× more than general repos while being pushed to less, in every one of the 20 usable quarters.

The AI boom shows up on GitHub as an attention boom first, and a code boom second.



CONCLUSION

AI/ML activity grew 2.35× from 2021 to 2026 vs 1.28× for general software, peaking at 2.26% in 2023-Q2 alongside GPT-4 — yet AI/ML is still under 1% of all public GitHub events. The durable change isn't volume but attention: AI/ML repos are starred 4.4× and forked 3.1× more than general repos while being pushed to less, in every usable quarter. The AI boom shows up on GitHub as an attention boom first, and a code boom second.

TOOLS USED

Phase 1 — Profiling (CLI, streaming): zcat, wc -c/wc -l, jq, grep -c, cut, awk, sort, uniq -c, column -t, parallelised with xargs -P 10, every command timed with time.

Phase 2 — Breaking (in-memory): Excel / Numbers row cap, pandas read_json under ulimit -v, peak RSS from /usr/bin/time -v.

Phase 3 — Scaling (distributed): PySpark on GCP Dataproc, raw data read straight from a GCS bucket via gs://..., clusters managed with gcloud dataproc.